Khoa Nguyen

Newark, New Jersey   nk569@njit.edu   (973)-978-6177   khoanguyen2410

RESEARCH INTEREST

My research focuses on building **reliable** and **privacy-preserving agentic systems** for real-world security and software environments. My work lies at the intersection of **LLM systems, tool-augmented agents, distributed learning, and differential privacy,** aim to enabling AI systems that can reason over complex data, interact with external tools, and operate safely in data-sensitive settings. My work centers on two directions:

- **Privacy-preserving LLM and agent systems:** bridging formal theoretical guarantees with empirical verification to design scalable defense mechanisms, utilizing local differential privacy, that protect prompts, latent states, and multi-step outputs against carefully designed reconstruction, extraction, and inference attacks under realistic adversarial settings.
- **Reliable agent deployment at industry scale:** translating foundational research into production-grade systems by leveraging distributed tuning and resource optimization to improve the scalability, controllability, and verified robustness of agentic workflows across security-critical software environments.

EDUCATION

New Jersey Institute of Technology — *Sep 2023 – Present*
Ph.D in Data Science
Advisor: Dr. NhatHai Phan

New Jersey Institute of Technology — *Sep 2019 – May 2023*
B.S. in Computer Science

INDUSTRY EXPERIENCE

OppyAI Inc. — *Newark, NJ*
Research Scientist — *Nov 2024 – Present*

- Helped launch OppyAI from one of my research projects, translating privacy-preserving AI research into a startup focused on real-world product and system development.
- Designed and developed core privacy-preserving techniques for LLMs, protecting sensitive data during training and inference while enabling scalable and reliable deployment.
- Built and maintained end-to-end data processing, training, and evaluation pipelines, scaling them across multi-node, multi-GPU environments and optimizing distributed workflows to improve efficiency, stability, and resource utilization for large-model development.

AT&T — *Bedminister, NJ*
Research Collaborator — *Nov 2025 – Present*

- **Project:** Machine Learning based Approach for Cellular Network Maintenance Scheduling.

TECHNICAL SKILLS

Frameworks: Pytorch, Huggingface, LLama-Factory, Verl, DeepSpeed, Ray.

Infrastructures & Tools: CUDA, Linux, Git, Docker, SQL.

Languages: Python, Bash, Java, C++, C.

RESEARCH PROJECTS

Privacy-Preserving Generation with LLMs — *Jan 2024 – Present*

- **Project Goals:** Develop privacy-preserving agentic systems that safeguard sensitive user information while preserving strong model utility, with a focus on reliable deployment in interactive, tool-augmented, and agent-based settings.
- **Methodology:** Developed NOIR, a split-learning LLM framework with local differential privacy (LDP); analyzed reconstruction risk through theoretical bounds on attack recovery; and extending the mechanism toward multimodal, tool-use, and agentic systems.

Program Structure-Aware LLM for Targeted Software Testing *Jan 2025 – Present*

- **Project Goals:** Enhance the reliability and controllability of agentic systems for software testing through targeted test-case generation aligned with specific program behaviors, supporting improved bug discovery, software assurance, and security analysis.
- **Methodology:** Designed a program structure-aware framework that conditions agentic systems on program representations to guide behavior-specific test generation, achieving higher precision and effectiveness compared to strong baseline methods.

FedChip: Federated LLM for AI Accelerator Chip Design *Apr 2024 – Apr 2025*

- **Project Goals:** Enable collaborative LLM-based hardware design across distributed participants without centralizing sensitive design data, improving both practicality and design quality in multi-party settings.
- **Methodology:** Developed a federated framework for AI accelerator chip design, constructed a multi-objective evaluation pipeline for design quality, and demonstrated substantial improvements over prior LLM-based hardware design methods.

Geographic-Aware Federated Learning *Sep 2023 – Apr 2024*

- **Project Goals:** Improve the scalability and effectiveness of federated learning in decentralized environments by modeling structured relationships among participants under realistic data heterogeneity.
- **Methodology:** Developed a geographic-aware federated learning framework that incorporates real-world relational structure across clients, and validated its effectiveness on large-scale distributed datasets.

PUBLICATIONS

1. ***Khoa Nguyen***, Khiem Ton, NhatHai Phan, Issa Khalil, Khang Tran, Cristian Borcea, Ruoming Jin, Abdallah Khreishah, My T Thai. *NOIR: Privacy-Preserving Generation of Code with Open-Source LLMs.* The USENIX Security Symposium (USENIX Security'26).
2. Khang Tran, ***Khoa Nguyen***, Cristian Borcea, NhatHai Phan. *Program Structure-aware Language Models: Targeted Software Testing beyond Textual Semantics.* Findings of the Association for Computational Linguistics (ACL'26).
3. ***Khoa Nguyen***, Khang Tran, NhatHai Phan, Cristian Borcea, Ruoming Jin, Issa Khalil. *SGFusion: Stochastic Geographic Gradient Fusion in Federated Learning.* International Conference on Big Data (BigData'25)
4. Mahmoud Nazzal, ***Khoa Nguyen***, Deepak Vungarala, Ramtin Zand, Shaahin Angizi, Hai Phan, Abdallah Khreishah. *FedChip: Federated LLM for Artificial Intelligence Accelerator Chip Design.* IEEE International Conference on LLM-Aided Design (ICLAD'25).

PATENTS

1. NhatHai Phan, ***Khoa Nguyen***, Khiem Ton, Issa Khalil, My Thai, Ruoming Jin. *System and Method for Private Generation of Code.* International Patent Application, No. PCT/US25/042391.

SERVICE

- **Teaching Assisstant at NJIT: DS110: Basic Foundations of AI, CS280: Programming Lang Concepts, DS677: Deep Learning, DS636: Data Analytics with R Program.**

ACHIVEMENTS

Winner – GfK's NextGen Data Science Hackathon *2022*

ICPC – Qualified for Greater NY Regional Contest *2022*

3rd Prize – Vietnam Science and Engineering Fair (ViSEF) *2017*